# 1. Project Planning & Management

## 1.1 Project Proposal

### Project Overview

**ECHO Egypt** is a smart tourism platform focused on helping users discover and experience Egypt based on their interests, moods, memories, and travel preferences.

The platform connects **places, experiences, real memories, AI-powered personalization, community contributions, and tourism partners** in one ecosystem.

The main concept is:

**Discover → Experience → Remember → Share → Contribute → Earn → Explore More**

### Objectives

The main objectives of ECHO Egypt are to:

- Help users discover Egyptian places and experiences.
- Provide personalized travel recommendations.
- Transform real user memories into possible experiences.
- Help users create personalized trip plans using AI.
- Allow users to share memories and useful information.
- Encourage meaningful community contributions (planned for later phases).
- Connect travelers with tourism businesses and local experiences (planned for later phases).

### Project Scope

The MVP scope includes:

- Experience and destination discovery.
- Mood-based discovery.
- Real memories and user-generated content.
- Memory → Experience.
- AI-powered experience planning based on mood, interests, budget, duration, and location.
- User accounts, profiles, and travel preferences.
- Maps and location-based features.
- Basic content management and moderation by administrators.

The following features belong to the long-term vision and are planned for later phases, outside the MVP:

- Advanced Travel DNA, personalized home, match percentages, and the post-experience learning loop.
- Community features: reviews, reports, points, badges, and the ECHO Passport.
- Tourism partners: business profiles, rewards, and business dashboards.
- Place Pulse, tourism intelligence, advanced AI, and predictive recommendations.
- Safety features such as check-in, emergency information, and SOS.

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# 1.2 Project Plan

## Timeline

| Activity | Start Date | End Date |
|---|---|---|
| Project Planning & Management | 2 Oct 2026 | 16 Oct 2026 |
| Literature Review | 9 Oct 2026 | 16 Oct 2026 |
| Requirements Gathering | 9 Oct 2026 | 25 Oct 2026 |
| System Analysis & Design | 9 Oct 2026 | 29 Oct 2026 |
| Implementation | 30 Oct 2026 | 30 Nov 2026 |
| Final Presentation, Testing & Reports | 30 Oct 2026 | 4 Dec 2026 |

# Gantt Chart

The chart below visualizes the activities, their overlaps, and their end dates (milestones).

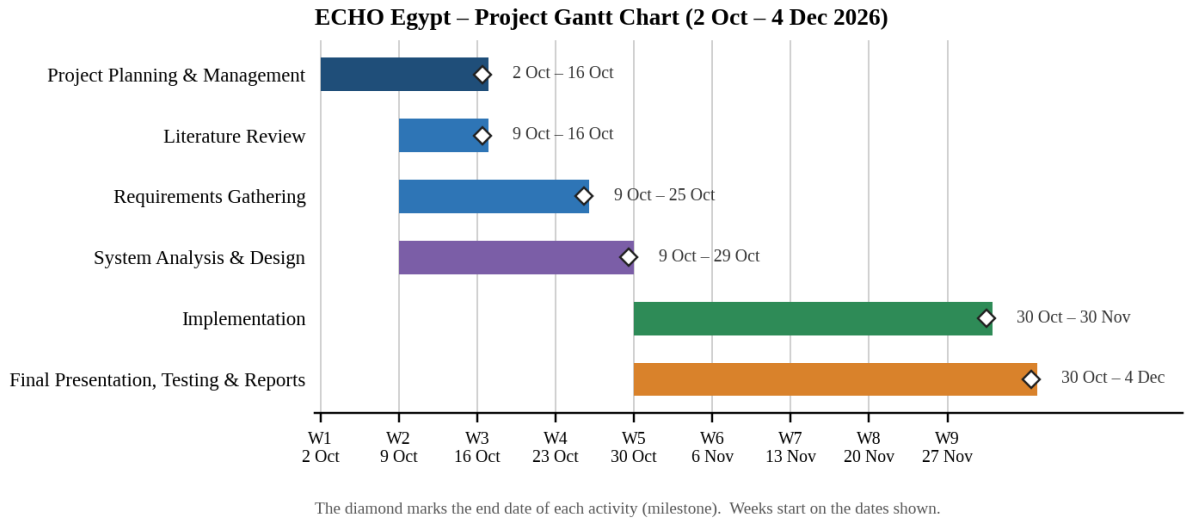



**Figure 1:** *Project Gantt chart based on the timeline above.*

# Main Milestones

1. Project planning completed (16 Oct 2026).
2. Project requirements identified (16 Oct 2026).
3. System analysis and design completed (6 Nov 2026).
4. MVP implementation completed (30 Nov 2026).
5. System testing completed (by 4 Dec 2026).
6. Documentation and final presentation completed (4 Dec 2026).

# Main Deliverables

- Project Proposal.
- Project Plan and Gantt Chart.
- Requirements Documentation.
- System Analysis & Design Documentation.
- Database and System Diagrams.
- UI/UX Designs.
- Implemented ECHO Egypt MVP.
- Source Code and README on GitHub.
- Deployed Application Link.
- Testing Documentation.
- API Documentation.

- User and Technical Documentation.
- Final Presentation.
- Project Infographic.

## Resources

The project will use:

- Team members for development, UI/UX, testing, and documentation.
- Git and GitHub for version control and project collaboration.
- Figma for UI/UX design.
- Next.js, React.js, TypeScript, and Tailwind CSS for development.
- MongoDB Atlas for database management.
- Cloudinary for media storage.
- AI/LLM APIs for AI-powered features.
- Map and location services for geographical features.

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# 1.3 Task Assignment & Roles

| Team Member | Main Responsibilities | Key Tasks (Proposed) |
|---|---|---|
| **Abdulaziz Arafa Abdulraouf Al-Saadani** | Frontend & Backend Development | Authentication, REST APIs, MongoDB models, AI planner integration, deployment |
| **Ibrahim Mohamed Ibrahim Fathallah** | UI/UX Design & Frontend Development | Figma designs, UI components, discovery and memory pages, usability testing |
| **Hassan Gamal Saad Rezk Obaed** | Frontend & Backend Development | Authentication, REST APIs, MongoDB models, AI planner integration, deployment |
| **Arwa Mohamed Salah Mohamed** | UI/UX Design & Frontend Development | Figma designs, UI components, discovery and memory pages, usability testing |

The team will collaborate on testing, documentation, integration, and final project preparation.

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# 1.4 Risk Assessment & Mitigation Plan

| Risk | Impact | Mitigation |
|---|---|---|
| Delay in development | High | Divide the project into smaller tasks and track progress regularly. |

| Risk | Impact | Mitigation |
|---|---|---|
| Integration problems | High | Define clear API interfaces and test components during development. |
| AI/API limitations | Medium | Use modular AI integration and provide fallback behavior where possible. |
| Database or data problems | High | Validate data and maintain regular backups during development. |
| UI/UX inconsistencies | Medium | Use shared Figma designs and common UI guidelines. |
| Team task delays | Medium | Clearly assign responsibilities and review progress regularly. |
| Bugs before final delivery | High | Perform continuous testing instead of waiting until the final stage. |
| Scope creep (the vision is larger than the MVP) | High | Limit implementation to the defined MVP scope and move other vision features to later phases. |
| Security or data protection issues | High | Use secure authentication, server-side validation, and keep API keys out of public code. |
| Deployment or hosting problems | Medium | Deploy early to a test environment and resolve deployment issues before the final week. |

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# 1.5 KPIs (Key Performance Indicators)

The following KPIs will be used to evaluate the project's success:

| KPI | Measurement | Target (Proposed) |
|---|---|---|
| **System Response Time** | Measure how quickly main pages and features respond. | 2 seconds or less for common API requests (excluding AI processing and large media transfers). |
| **System Availability** | Measure whether the deployed system remains accessible. | At least 99% availability during the evaluation and demonstration period. |
| **Feature Completion** | Percentage of planned MVP features successfully implemented. | 100% of the core MVP features. |
| **Bug Rate** | Number of detected and unresolved bugs during testing. | No unresolved critical bugs at final delivery. |
| **AI Recommendation Relevance** | Evaluate whether generated recommendations match user inputs such as mood, interests, budget, and duration. | At least 80% of tested plans judged relevant to the given inputs. |
| **User Task Completion** | Percentage of users able to complete key flows such as discovering an experience and creating a journey. | At least 80% of test users complete the key flows. |

| KPI | Measurement | Target (Proposed) |
|---|---|---|
| **User Engagement** | Measure interactions such as experiences viewed, memories created, and contributions submitted. | Tracked during testing; no fixed target for the MVP. |
| **Project Delivery** | Completion of planned deliverables within the project timeline. | 100% of planned deliverables submitted by the deadlines. |

**Note:** The targets are proposed values for the MVP and should be confirmed by the team and the lecturer.